

Chapter 6

International Parity Conditions

Multinational Business Finance – Eiteman et al.

Exchange Rates • Inflation • Interest Rates • Forward Markets

Learning Objectives



Prices & Exchange Rates

How price levels and inflation determine exchange rates



Interest Rate Effects

How interest rates reflect inflationary forces within each currency



Forward Market Expectations

How forward markets reflect future spot rate expectations



Equilibrium Conditions

How spot/forward markets align with interest & inflation differentials

01

Prices & Exchange Rates

Law of One Price • Purchasing Power Parity • Pass-Through

The FX Quotation System Summary Table

Notation Style	Conceptual System	Algebraic / Market Meaning	How to Explain to Students
USD/VND = 26,000	FX Market Convention	Base / Quote structure. The first currency is the base and always equals 1 unit .	1 USD can be exchanged for 26,000 VND .
S = 26,000 VND/USD	Algebraic / Unit Convention	The slash / means " per ".	There are 26,000 VND per 1 USD .
VND 26,000/USD	Commercial / Daily Style	The slash / means "per".	The price of 1 USD is 26,000 VND .
S _{VND/USD} = 26,000	Academic / Textbook Style	The slash / means "per".	The price of 1 USD is 26,000 VND.

International Finance Theories Summary

Theory	Formula	What it Relates	Japan Example
1. Purchasing Power Parity (PPP)	$\% \Delta S \approx \pi_{US} - \pi_{JP}$	Inflation ↔ Exchange Rate	Japan inflation 1%, US inflation 5% → Yen appreciates ≈ 4%
2. Fisher Effect (FE)	$i = r + \pi$	Inflation ↔ Interest Rates	Inflation 1% lower → interest rate ≈ 1% lower
3. International Fisher Effect (IFE)	$\% \Delta S \approx i_{US} - i_{JP}$	Interest Rates ↔ Exchange Rate	US rate 6%, Japan rate 2% → Yen appreciates ≈ 4%
4. Interest Rate Parity (IRP)	$F = S \times \frac{1+i_f}{1+i_h}$	Interest Rates ↔ Forward Rate	Japan rate lower → Yen at forward premium
5. Forward Rate as Unbiased Predictor (FRUP)	$\% \Delta S \approx \frac{F-S}{S}$	Forward Premium ↔ Future Spot Change	4% forward premium → Yen expected to appreciate 4%

Law of One Price (LOP)

KEY

The Core Idea

1. Identical product sold in two markets
2. No restrictions on the sale
3. Equal transportation costs

⇒ Then the price should be the same in both markets.

$$P_{\$} \times S = P_{¥}$$

$P_{\$}$ = price in USD | S = spot rate | $P_{¥}$ = price in Yen

☀ Real-World Example: Big Mac Index

The Economist's Big Mac Index tests LOP globally.

Example:

- Big Mac costs \$5.58 in the US
- Big Mac costs ¥700 in Japan
- Implied exchange rate: $¥700 \div \$5.58 \approx ¥125.45/\$$

If actual rate is ¥130/\$, the yen is **undervalued** vs. the USD.

Purchasing Power Parity (PPP) – Absolute Version

KEY

Absolute PPP extends LOP to ALL goods & services.

If markets are efficient, the 'real' PPP exchange rate can be found by comparing identical product prices in two currencies.

How It Works

1. Take any identical good (e.g. a laptop)
 2. Compare its price in USD vs EUR
 3. Compute the implied exchange rate
 4. Compare to the market spot rate
- Difference = currency over/undervaluation

Limitations

- Many goods are non-tradeable (haircuts, rent)
- Transportation costs distort prices
- Trade barriers (tariffs, quotas) exist
- Works better for traded goods only
- Does not predict short-term exchange rates

Relative PPP (RPPP)

KEY

"If the spot exchange rate starts in equilibrium, any change in the differential rate of inflation tends to be offset by an equal but opposite change in the spot exchange rate — over the long run."

RPPP Formula (approx.):

$$\% \Delta S \approx \pi_f - \pi_d$$

Empirical Findings:

- ✓ Works well over the **very long run** (10+ years)
- ✓ Better for high-inflation countries
- ✗ Poor predictor for short-term rates
- ✗ Fails in developed, low-inflation markets



Numeric Example

US inflation: 2% | Brazil inflation: 6%

Expected BRL depreciation:

$$\% \Delta S \approx 6\% - 2\% = 4\%$$

So BRL should weaken ~4% vs. USD.

Practice: Purchasing Power Parity

PRACTICE

Questions

- Q1.** A coffee costs \$4.00 in the US and €3.60 in Germany. What does LOP imply for the EUR/USD rate?
- Q2.** Given GBP/USD 1.35. If US inflation is 3% and UK inflation is 7%, what is the expected change in the GBP/USD rate per RPPP?
- Q3.** In which two situations does PPP work best empirically? Why?

Exchange Rate Pass-Through

KEY

Measures how much a change in exchange rates affects the prices of imported/exported goods.

Complete Pass-Through

100% of exchange rate change is reflected in import prices.
Exporter keeps margins intact.

Partial Pass-Through

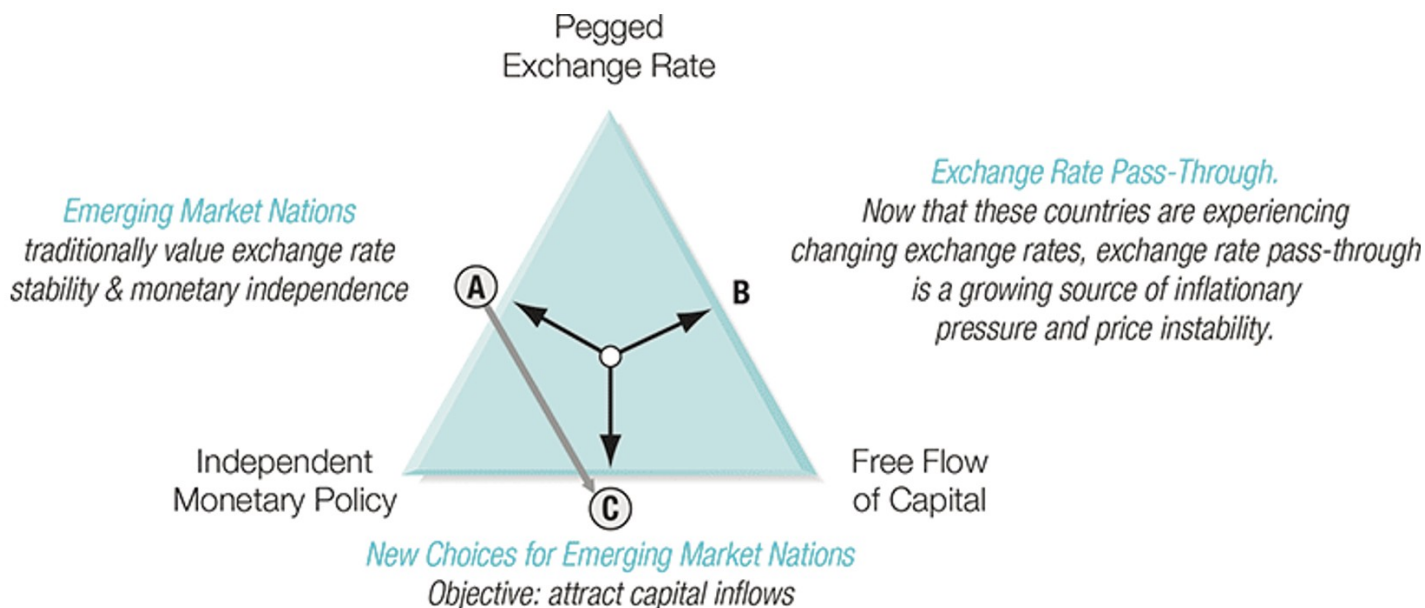
Only part of the change is passed to buyers.
Exporter absorbs some via margin compression.

Zero Pass-Through

Prices do not change at all.
Exporter fully absorbs currency move.



Exhibit 6.4 Pass-Through, the Impossible Trinity, and Emerging Markets



Many emerging market countries have chosen to move from Point A to Point C, exchanging fixed exchange rates for the chance of attracting capital inflows. The result is that these countries are now the subject to varying levels of exchange rate pass-through.



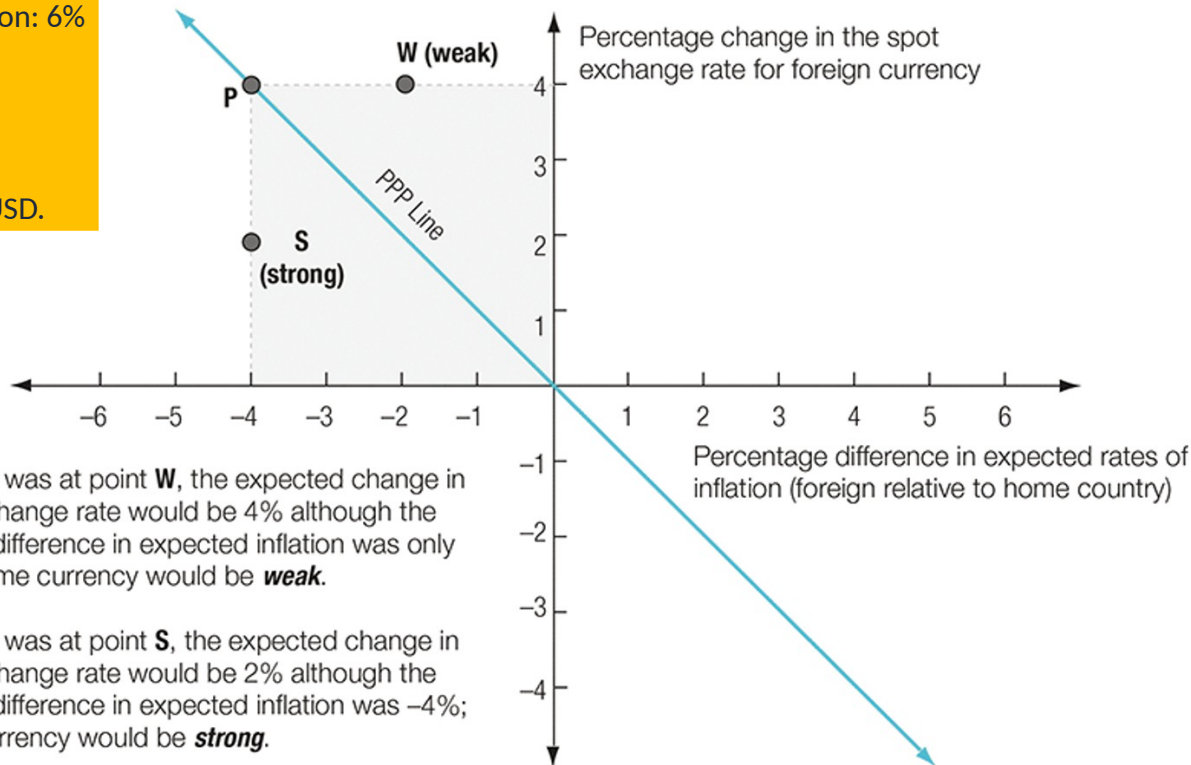
Exhibit 6.2 Relative Purchasing Power Parity (PPP)

US inflation: 2% | Brazil inflation: 6%

Expected BRL depreciation:

$$\% \Delta S \approx 6\% - 2\% = 4\%$$

So BRL should weaken ~4% vs. USD.



If the market was at point **W**, the expected change in the spot exchange rate would be 4% although the percentage difference in expected inflation was only -2%; the home currency would be **weak**.

If the market was at point **S**, the expected change in the spot exchange rate would be 2% although the percentage difference in expected inflation was -4%; the home currency would be **strong**.

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2

Interest Rates & Exchange Rates

Fisher Effect • International Fisher Effect

The Fisher Effect

KEY

Nominal interest rates in each country = required real rate of return + expected inflation compensation.

$$i \approx r + \pi$$

i = nominal rate r = real rate π = expected inflation

Empirical Finding:

The Fisher effect holds reasonably well for short-maturity government securities (T-bills/notes), tested using ex-post inflation rates.



Numeric Example

Real rate (r) = 2%

Expected inflation (π) = 4%

Nominal rate (i) = 2% + 4% = 6%

→ If inflation rises to 7%,
nominal rate rises to ~9%.

Next: The International Fisher Effect extends this across borders →

International Fisher Effect (IFE) – 'Fisher Open'

KEY

The spot exchange rate should change in an equal but opposite amount to the difference in interest rates between two countries.

$$(S_1 - S_0) / S_0 \approx i_{\$} - i_{¥}$$

S_0 = current spot | S_1 = future spot | $i_{\$}$ = USD rate | $i_{¥}$ = JPY rate



Intuition

If US rates are 5% and Japan rates are 1%, investors will rush to USD assets.

This demand bids up the USD until the expected depreciation of USD erases the advantage → equilibrium.



Numeric Example

$i_{\$} = 5\%$, $i_{¥} = 1\%$

Current spot: ¥130/\$

IFE predicts USD will depreciate by ~4%

Future spot $\approx$ ¥130 $\times$ (1 - 0.04) $\approx$ ¥124.8/\$

(Yen appreciates vs. Dollar)

Practice: Fisher Effect & IFE

PRACTICE

Questions

- Q1.** If the real rate is 2.5% and expected inflation is 3.5%, what is the nominal interest rate per the Fisher Effect?
- Q2.** Australia has a nominal interest rate of 6% and the US rate is 3%. If the current AUD/USD rate is 0.72, what does the IFE predict for the future spot rate?
- Q3.** Why do investors borrow in low-interest countries and invest in high-interest countries? What risk does this carry?

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Forward Rates & Interest Rate Parity

Forward Rate Calculation · IRP · Covered vs. Uncovered Arbitrage

The Forward Rate

KEY

A forward rate is an exchange rate quoted today for settlement at a specific future date.

$$F_{90}^{\text{SF}/\$} = S^{\text{SF}/\$} \times \frac{\left[1 + \left(i^{\text{SF}} \times \frac{90}{360} \right) \right]}{\left[1 + \left(i^{\$} \times \frac{90}{360} \right) \right]}$$



Numeric Example

$S = 0.9500 \text{ SF}/\$$

$i_{\text{SF}} (90\text{d}) = 2\% \text{ ann.} \rightarrow 0.50\% \text{ for } 90\text{d}$

$i_{\$} (90\text{d}) = 4\% \text{ ann.} \rightarrow 1.00\% \text{ for } 90\text{d}$

$F = 0.9500 \times (1.005 / 1.010) \approx 0.9453$

$\rightarrow \text{SF at a premium vs. } \$$

Forward Premium / Discount

$$f^{\text{SF}} = \frac{\text{Spot} - \text{Forward}}{\text{Forward}} \times \frac{360}{\text{Days}} \times 100.$$

Positive = foreign currency at PREMIUM

Negative = foreign currency at DISCOUNT

Annualised for easy comparison.

Interest Rate Parity (IRP)

KEY

The difference in national interest rates for similar-risk securities should equal — but be opposite in sign to — the forward rate premium or discount.

$$F = S \times \frac{1+r_d}{1+r_f}$$

IRP In Words

High-interest currency trades at a
FORWARD DISCOUNT.

Low-interest currency trades at a
FORWARD PREMIUM.

This prevents free arbitrage profit.



Numeric Example

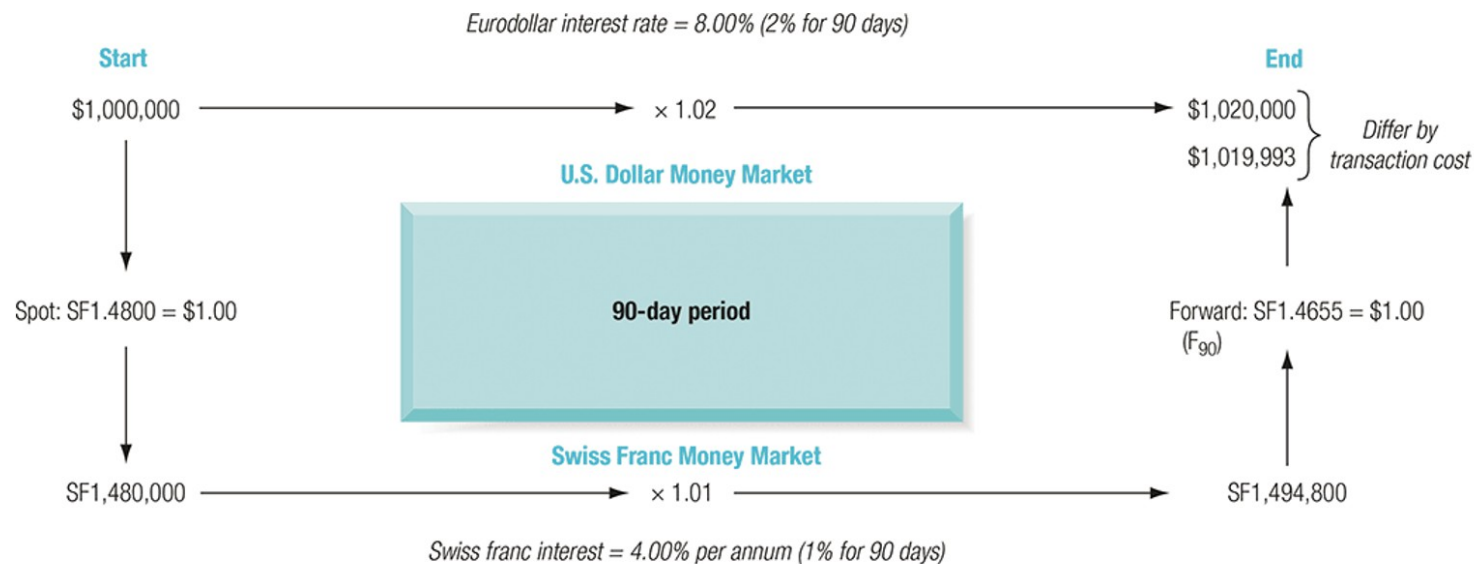
US rate: 5% | Germany rate: 2%
Spot: 1.10 USD/EUR

IRP forward rate $\approx 1.10 \times (1.05/1.02)$
 $\approx 1.10 \times 1.0294 \approx 1.132$ USD/EUR

EUR trades at ~3% forward premium.



Exhibit 6.6 Interest Rate Parity (IRP)



Covered Interest Arbitrage (CIA)

KEY

When IRP doesn't hold, a risk-free profit opportunity exists. Traders exploit it — covered by forward contracts.

1 Borrow

Borrow in the lower-interest currency (e.g. SW at 1%)

2 Convert

Convert to higher-interest currency at spot rate

3 Invest

Invest at the higher rate (e.g. USD at 2%) for the same period

4 Lock Rate

Sell the USD forward at today's forward rate

5 Profit

Receive foreign return, repay SW loan → keep spread as risk-free profit

⚡ Arbitrage forces IRP back to equilibrium — the window closes quickly in efficient markets.

Covered vs Uncovered Interest Rate Parity

KEY

Imagine the US (Home) interest rate is 5% and the Japan (Foreign) rate is 1%. The current spot rate is 100.00 JPY/USD ($\frac{\text{JPY}}{\text{USD}}$).

Under CIRP (The Arbitrage View):

An investor wants to ensure they don't lose money. The theory states that because the US offers a 4% higher return, the market forward rate *must* adjust downward so nobody can abuse this gap.

The 1-year forward rate automatically locks in at a discount:

$$F = 100 \times \frac{1 + 0.01}{1 + 0.05} \approx \mathbf{96.19 \text{ JPY/USD}}$$

If an investor tries to move money to the US to chase the 5%, locking in the forward contract at 96.19 ensures their net return when converting back to Yen is exactly 1%—wiping out the premium.

Under UCIRP (The Expectations View):

Instead of looking at contracts, UCIRP looks at economic fundamentals. It predicts that because the US has a higher nominal interest rate (often due to higher expected inflation), market forces will cause the US Dollar to **depreciate** over the next year by roughly 4%.

Therefore, the market *expects* the actual spot rate in one year ($E(S_t)$) to drift down to **96.19 JPY/USD**.

Covered vs Uncovered Interest Rate Parity

KEY

Why Does UCIRP Fail in the Real World? (The Carry Trade)

While CIRP is bound by tight mathematical arbitrage in modern trading desks, UCIRP fails empirically on a regular basis in the short-to-medium term. According to UCIRP, high-interest currencies should depreciate. In reality, international investors routinely chase high yields. They borrow in low-interest currencies (like JPY) and flood into high-interest currencies, actually causing the high-yielding currency to appreciate in the short term due to high demand. This market anomaly is known as the FX Carry Trade, and it represents a direct, ongoing violation of Uncovered Interest Rate Parity.

Uncovered Interest Arbitrage (UIA) – Yen Carry Trade

KEY

Feature	Covered (CIA)	Uncovered (UIA)
Forward contract used?	✓ Yes — locks in rate	✗ No — exposed to spot rate
Currency risk?	None	High — spot rate may move against you
Expected return?	Small, risk-free	Large, but risky
Example	Arbitrage locked via hedge	Yen Carry Trade (borrow JPY, invest USD)



Yen Carry Trade Example

1. Borrow ¥100M at 0.5% (Japan rate)
2. Convert to USD at spot ¥130/\$ → get ~\$769,230
3. Invest \$769,230 in US at 5% for 1 year → ~\$807,692
4. Convert back at end-of-year spot rate
 - If spot stays ¥130/\$: repay ¥100.5M, pocket ~¥5M profit ✓
 - If yen strengthens to ¥120/\$: repay ¥96.9M — big loss! ✗

Practice: IRP, CIA & UIA

PRACTICE

Questions

- Q1.** USD rate = 4%, EUR rate = 1%. Spot rate = 1.10 USD/EUR. What is the IRP-implied 1-year forward rate?
- Q2.** You notice the actual forward rate is 1.08 USD/EUR. Is there a CIA opportunity? If so, describe the steps (borrow EUR or USD?). Calculate the actual profit.
- Q3.** Why is UIA riskier than CIA? Give one scenario where UIA leads to a loss.

Covered vs Uncovered Interest Rate Parity

KEY

Feature	Covered Interest Rate Parity (CIRP)	Uncovered Interest Rate Parity (UCIRP)
Core Equation	$F = S \times \frac{1+i_{\text{home}}}{1+i_{\text{foreign}}}$	$E(S_t) = S \times \frac{1+i_{\text{home}}}{1+i_{\text{foreign}}}$
Key Exchange Rate	Uses the Forward Rate (F)	Uses the Expected Future Spot Rate ($E(S_t)$)
Risk Mitigation	Hedged/Covered via forward contracts	Unhedged/Exposed to market fluctuations
Driving Force	Risk-free arbitrage	Investor expectations and market efficiency
Empirical Validity	Holds true almost perfectly (except during severe liquidity crises like 2008).	Fails frequently in the short run (drives the famous "Carry Trade" strategy).
Time Horizon	Short to medium term (matching forward market liquidity).	Long-term economic equilibrium.

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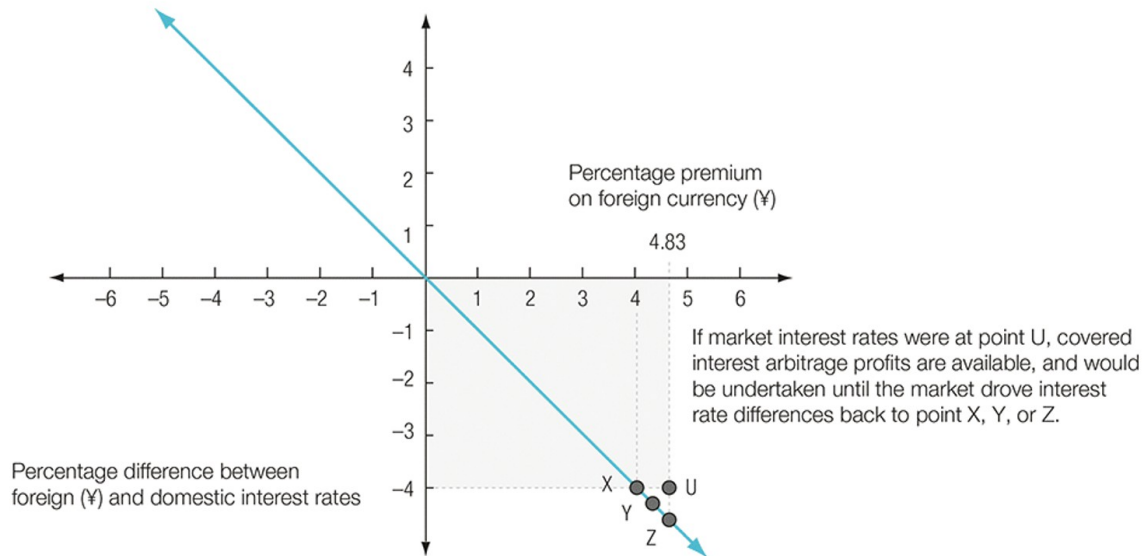
Equilibrium & The Big Picture

IRP Equilibrium · Forward as Predictor · Parity Conditions Map

IRP Equilibrium: Point U and Arbitrage Forces

KEY

When the market is off the IRP line (disequilibrium), arbitrageurs move it back. The process is virtually risk-free and happens fast.



How Equilibrium Is Restored

1. Point U = off the IRP line
2. All investors see the same arb opportunity
3. CIA trades start immediately:
 - Borrow in cheap currency
 - Invest in high-yield currency
 - Sell forward to cover risk
4. Activity drives rates back to IRP line
5. Profit disappears → equilibrium restored

⚡ In practice, electronic trading means arbitrage windows last milliseconds — markets are highly efficient.

Forward Rate as an Unbiased Predictor

KEY

Some forecasters believe the forward rate is the best available predictor of the future spot rate.

What 'Unbiased' Means

The forward rate will, on average:

- Overestimate the future spot rate as often as it underestimates it

Distribution of future spot rates is centered on the forward rate.

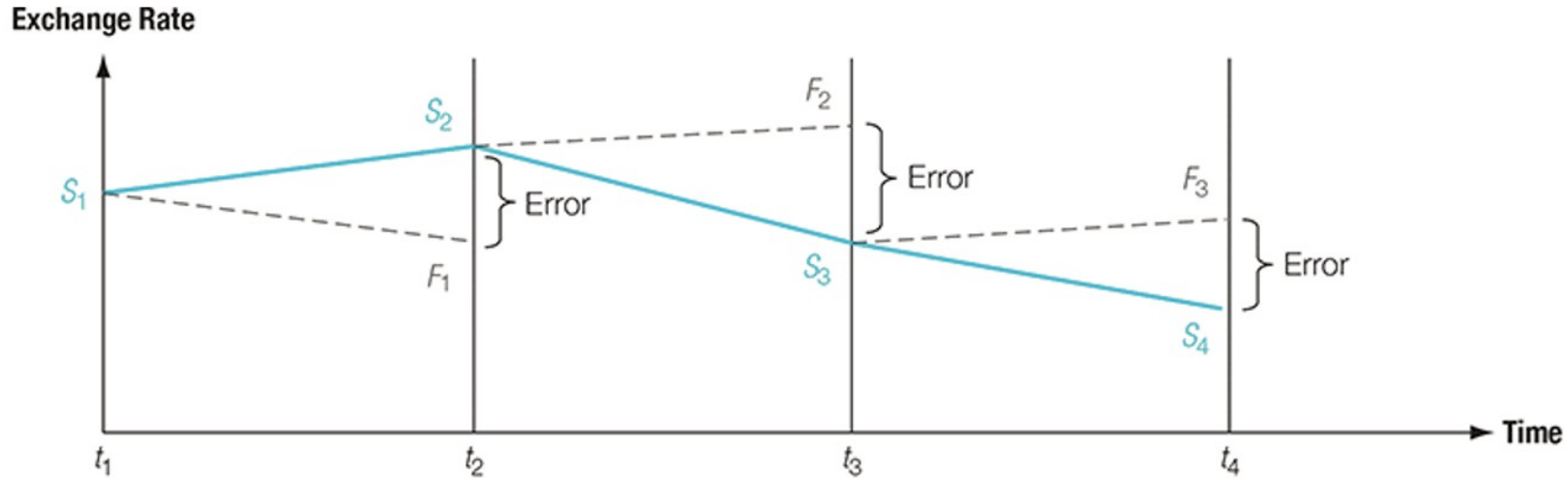
Not perfect — but no systematic bias.

Implications

- Speculators cannot consistently profit from forward-spot differences alone
- The forward rate is the market's consensus forecast
- Actual spot $\neq$ forward due to risk premiums and new information

Quick Check: If today's 90-day forward rate for GBP is 1.25 USD/GBP, the unbiased predictor theory says the expected future spot in 90 days is **1.25 USD/GBP**. It will often miss, but on average it won't be systematically above or below.

Exhibit 6.10 Forward Rate as an Unbiased Predictor of Future Spot

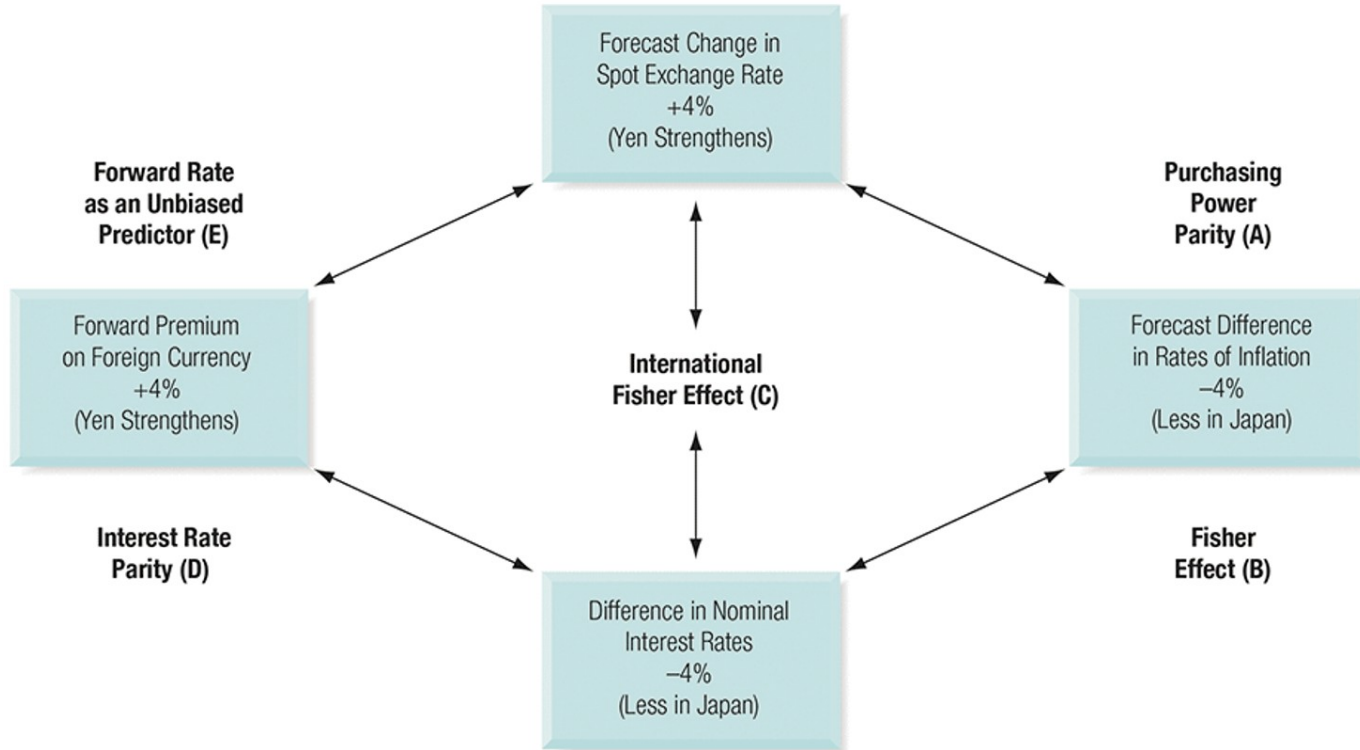


The forward rate available “today” (t) for delivery at a future time ($t + 1$) is used as a forecast or *predictor* of the spot rate at time $t + 1$. The difference between the spot rate which then occurs and the forward rate is the forecast error. When the forward rate is termed an “unbiased predictor of the future spot rate” it means that the errors are normally distributed around the mean future spot rate (the sum of the errors equal zero).

Int'l Parity Conditions – Big Picture Summary

KEY

All five parity conditions link together in equilibrium:



International Finance Theories Summary

Theory	Formula	What it Relates	Japan Example
1. Purchasing Power Parity (PPP)	$\% \Delta S \approx \pi_{US} - \pi_{JP}$	Inflation $\leftrightarrow$ Exchange Rate	Japan inflation 1%, US inflation 5% $\rightarrow$ Yen appreciates $\approx$ 4%
2. Fisher Effect (FE)	$i = r + \pi$	Inflation $\leftrightarrow$ Interest Rates	Inflation 1% lower $\rightarrow$ interest rate $\approx$ 1% lower
3. International Fisher Effect (IFE)	$\% \Delta S \approx i_{US} - i_{JP}$	Interest Rates $\leftrightarrow$ Exchange Rate	US rate 6%, Japan rate 2% $\rightarrow$ Yen appreciates $\approx$ 4%
4. Interest Rate Parity (IRP)	$F = S \times \frac{1+i_f}{1+i_h}$	Interest Rates $\leftrightarrow$ Forward Rate	Japan rate lower $\rightarrow$ Yen at forward premium
5. Forward Rate as Unbiased Predictor (FRUP)	$\% \Delta S \approx \frac{F-S}{S}$	Forward Premium $\leftrightarrow$ Future Spot Change	4% forward premium $\rightarrow$ Yen expected to appreciate 4%

Comprehensive Practice: All Parity Conditions

PRACTICE

Work through this scenario applying **ALL** the conditions you've learned:

Scenario: The spot rate is USD 1.30/GBP. US nominal interest rate is 5%; UK nominal rate is 3%. US inflation is 2%; UK inflation is 4%.

Q1 – PPP

What does RPPP predict for the future GBP/USD rate?

Q2 – Fisher Effect

If UK real rate = 1%, what's the UK real inflation expectation? Is the Fisher Effect satisfied?

Q3 – IFE

What does the IFE predict for the GBP/USD exchange rate in 1 year?

Q4 – IRP

What is the IRP-implied 1-year forward rate for GBP?

Q5 – Synthesis

If the actual forward rate is 1.26, is there a CIA opportunity? If so, which direction?

Key Takeaways – Chapter 6

 LOP → Absolute PPP: identical goods should cost the same when converted to a common currency.

 Relative PPP: high-inflation currencies depreciate; works best long-run and in high-inflation economies.

 Fisher Effect: nominal rate = real rate + inflation. IFE: interest differentials $\approx$ expected currency change.

 IRP: forward premium/discount exactly offsets interest differentials — no free CIA profit in equilibrium.

 CIA vs UIA: CIA is hedged (risk-free); UIA (carry trade) is not — currency risk can wipe gains.

 All five conditions connect: inflation $\leftrightarrow$ exchange rates $\leftrightarrow$ interest rates $\leftrightarrow$ forward rates.